

Comparative Study on Friction Compensation using Coulomb and Dahl Models with Extended and Unscented Kalman Filters

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Abstract—In this paper, the Coulomb and the Dahl friction models have been simulated and estimated using extended Kalman filter (EKF) and unscented Kalman filter (UKF). The estimated friction is used by a controller to compensate and track a desired sinusoidal position profile. In the simulation, the real friction is estimated using EKF and UKF with the Coulomb and the Dahl friction models. The designed controller consists of both feedback and feedforward for position tracking. The proportional and derivative (PD) output feedback is selected. The desired trajectory inertia and the estimated friction are compensated as the feedforward. The tracking performance is analyzed in terms of the normalized root mean square error. It is found that the UKF based controller using Dahl friction compensation gives the lowest sinusoidal position tracking error, outperforming the EKF based controller counterpart.

Index Terms—Unscented Kalman filter, Extended Kalman filter, PD controller, Dahl model friction.

I. INTRODUCTION

Friction is a phenomena occurring in moving systems and can cause tracking errors, especially in precision motion control systems, such as robotic surgery and hard disk drive head position control [1], [2], [3]. Several control techniques have been proposed to compensate for the friction in systems, for example, [4], [5], [6], [7]. In the model-based control design, usually the friction, as a state or an output of a dynamical model, is estimated. The high level of complexity in the friction physics underlies the nonlinearity level of the friction model. The estimated friction is then used in the controller to cancel or to compensate for the actual friction to better regulate the trajectory of the output.

In several applications, the classical static Coulomb friction model is widely used because of its relatively simple structure, and the information is adequate for the actual friction compensation in the system with infrequent mode changes. However, when the friction has sizable magnitude and periodically undergoes stick/slip mode switchings, an accurate friction model is vital to achieve the accurate compensation and optimal or suboptimal tracking performance. In this case, the Coulomb friction model is not sufficient to capture physical behavior of the actual friction, especially when velocity is in the vicinity of zero. Therefore, dynamic friction models, for instance, Dahl [8] and LuGre [9] models have been proposed and shown to be more beneficial.

It was found that the dynamical friction models appeared nonlinear, complex and some parameters could not be easily measured. To receive all these parameters, an observed-based technique can be employed. A simple method to identify these parameters is an off-line identification; however, using this technique, all model parameters are used as fixed known values during the estimation process. Other solution is obtained via the on-line friction estimation [10], [11]. Using these methods, the model parameters are updated and can result in lower estimation error and more accurate tracking performance.

In on-line estimation, besides the unknown friction, there are measurement noises, several unknown parameters in the presence of the process noise which is inherited from the process itself. The Kalman filter (KF) technique is a well known on-line estimation technique, successfully adopted in many applications. For nonlinear systems, an extended Kalman filter (EKF) and an unscented Kalman filter (UKF), which are shown to be more precise, should be applied.

Previously, KF techniques have been applied for friction estimation applications. [12], [13], and [14] have proposed friction estimation using extended Kalman-Bucy filter (EKBF) and found that the friction estimation using EKBF yielded satisfying performance. In [15], both EKF and UKF with the Coulomb friction model were used to estimate the Coulomb friction parameter. Therein, the real friction was Coulomb.

In this paper, both EKF and UKF with Coulomb (static) and Dahl (dynamic) friction models are studied and used to estimate the real friction which is assumed to be of Dahl model. A PD output feedback controller is designed and superimposed with the estimated friction and the desired path inertia terms as the feedforward. The rest of paper is organized as follows. In the next section, a brief detail of friction model is presented and followed by the algorithms of EKF, UKF and the controller design. Simulation implementation is given in Section IV and conclusion is presented in the last section.

II. FRICTION MODEL

In this study, a simple rotational dynamic system equation is considered and given in (1)

$$J\ddot{\theta} = u - T_f \quad (1)$$

where J is the moment of inertia ($\text{kg}\cdot\text{m}^2$), $\ddot{\theta}$ is the rotational acceleration (rad/sec^2), u is the input torque ($\text{N}\cdot\text{m}$) and T_f is the friction torque ($\text{N}\cdot\text{m}$).

A. Coulomb Friction Model

A static Coulomb friction is widely used and easy to implement. The Coulomb friction model is presented in (2)

$$T_f = T_c \text{sgn}(\dot{\theta}) \quad (2)$$

where T_c is the magnitude of the Coulomb friction and $\dot{\theta}$ is the rotational velocity (rad/sec). It has been found that the Coulomb friction model in (2) cannot capture accurate friction behavior near zero velocity as well as being discontinuous [16]. Many models have been proposed in previous years; however, in this paper, Dahl model is the only real dynamic friction considered.

B. Dahl Friction Model

Based on the derivative $\frac{dT_f}{d\dot{\theta}}$ in (3), the dynamic friction of the Dahl model is depicted in (4).

$$\frac{dT_f}{d\dot{\theta}} = \sigma \left(1 - \frac{T_f}{T_c} \text{sgn}(\dot{\theta}) \right)^\delta \quad (3)$$

$$\frac{dT_f}{dt} = \frac{dT_f}{d\dot{\theta}} \frac{d\dot{\theta}}{dt} = \frac{dT_f}{d\dot{\theta}} \dot{\theta} = \sigma \left(1 - \frac{T_f}{T_c} \text{sgn}(\dot{\theta}) \right)^\delta \dot{\theta} \quad (4)$$

where σ and δ are constant parameters. In this paper, δ is set to 1, therefore (4) is rewritten as (5)

$$\frac{dT_f}{dt} = \sigma \left(1 - \frac{T_f}{T_c} \text{sgn}(\dot{\theta}) \right) \dot{\theta} \quad (5)$$

III. KALMAN FILTER (KF)

Kalman filter was first proposed by [17] for solving discrete time linear filtering problems. The basic operations of KF can be divided in two parts: time update and measurement update. In linear systems where all parameters are known, KF is an optimal whereas for nonlinear systems, extended Kalman filter (EKF) and unscented Kalman filter (UKF) are preferred. The EKF algorithm is based on the linearization of the estimated state of a nonlinear system which leads EKF to be suboptimal and sensitive for divergent problem. To deal with this problem, UKF was proposed by [18] and it operates based on a set of sigma points. The set is subject to the unscented transform (UT) based on the mean and covariance of nonlinear system.

The basic algorithms of EKF and UKF are presented as follows.

A. Extended Kalman Filter (EKF)

Consider a nonlinear system process equation and a measurement equation in (6) and (7), respectively.

$$x_{k+1} = f(x_k, u_k) + w_k \quad (6)$$

$$z_k = h(x_k, u_k) + v_k \quad (7)$$

where x_k is the dimensional state vector. u_k is the dimensional input vector at time k . w_k is the dimensional process white Gaussian noise vector with zero mean and covariance is M .

z_k is the dimensional measurement vector. v_k is the dimensional measurement white Gaussian noise with zero mean and covariance is R . w_k and v_k are independent.

The EKF algorithm can be computed as follows:

- state estimate and state covariance matrix: $\hat{x}_{k|k}, P_{k|k}$
- state prediction: $\hat{x}_{k+1|k} = f(\hat{x}_{k|k}, u_k)$
- state prediction covariance matrix:
 $P_{k+1|k} = F_k P_{k|k} F_k^T + M$
 $(F_k \text{ is the Jacobian matrix of } f(\hat{x}_{k|k}, u_k))$
- measurement prediction: $\hat{z}_{k+1|k} = h(\hat{x}_{k+1|k}, u_{k+1})$
- measurement prediction covariance:
 $S_{k+1} = H_{k+1} P_{k+1|k} H_{k+1}^T + R$
 $(H_{k+1} \text{ is the Jacobian matrix of } h(\hat{x}_{k+1|k}, u_{k+1}))$
- filter gain: $K_{k+1} = P_{k+1|k} H_{k+1}^T S_{k+1}^{-1}$
- updated state estimate:
 $\hat{x}_{k+1|k+1} = \hat{x}_{k+1|k} + K_{k+1}(z_{k+1} - \hat{z}_{k+1|k})$
- updated state covariance matrix:
 $P_{k+1|k+1} = P_{k+1|k} - K_{k+1} S_{k+1} K_{k+1}^T$

B. Unscented Kalman Filter (UKF)

The algorithm of UKF (for additive noise case) [19] are computed as follows:

- state estimate and state covariance matrix: $\hat{x}_{k|k}, P_{k|k}$
- weight matrices:
 $W_o^m = \frac{\lambda}{(N_x + \lambda)}$
 $W_o^c = \frac{\lambda}{(N_x + \lambda)} + (1 - \alpha^2 + \beta)$
 $W_i^m = W_i^c = \frac{1}{2(N_x + \lambda)}$
 where $i = 1, \dots, 2N_x$, and
 the scaling parameter, $\lambda = \alpha^2(N_x + \kappa) - N_x$.
 The constant α is set as a small positive value, e.g. $[10^{-4}, 1]$.
 The constant κ is usually set at 0 or $3 - N_x$, and
 $\beta = 2$ is optimal for the Gaussian distribution.
- a set of sigma points:
 $\mathcal{X}_k = [x_1 \ x_2 \ x_3]$
 where $x_1 = \hat{x}_k, x_2 = \hat{x}_k + \sqrt{(N_x + \lambda)P_k}$, and
 $x_3 = \hat{x}_k - \sqrt{(N_x + \lambda)P_k}$
- state prediction: $\hat{\mathcal{X}}_{k+1|k} = f(\mathcal{X}_k, u_k)$
 $\hat{x}_{k+1|k} = \sum_{i=0}^{2N_x} W_i^m \hat{\mathcal{X}}_{i,k+1|k}$
- state prediction covariance matrix:
 $P_{k+1|k} = \sum_{i=0}^{2N_x} W_i^c \mathbf{X} \mathbf{X}^T + M$
 where $\mathbf{X} = \mathcal{X}_{i,k+1|k} - \hat{x}_{k+1|k}$.
- $\mathcal{X}_{k+1|k} = [x_{1k} \ x_{2k} \ x_{3k}]$
 where $x_{1k} = \hat{x}_{k+1|k}$,
 $x_{2k} = \hat{x}_{k+1|k} + \sqrt{(N_x + \lambda)P_{k+1|k}}$, and
 $x_{3k} = \hat{x}_{k+1|k} - \sqrt{(N_x + \lambda)P_{k+1|k}}$
- measurement prediction:
 $\hat{z}_{k+1|k} = h(\mathcal{X}_{k+1|k}, u_{k+1})$
 $\hat{z}_{k+1|k} = \sum_{i=0}^{2N_x} W_i^m \mathcal{Z}_{i,k+1|k}$
- measurement prediction covariance:
 $S_{k+1} = \sum_{i=0}^{2N_x} W_i^c \mathbf{Z} \mathbf{Z}^T + R$
 where $\mathbf{Z} = \mathcal{Z}_{i,k+1|k} - \hat{z}_{k+1|k}$.
- cross covariance: $P_{xz,k+1} = \sum_{i=0}^{2N_x} W_i^c \mathbf{X} \mathbf{Z}^T$
- filter gain: $K_{k+1} = P_{xz,k+1} S_{k+1}^{-1}$

TABLE I
DAHL MODEL PARAMETERS [16]

parameters	values
J	0.0023
σ	260
T_c	0.28

- updated state estimate:

$$\hat{x}_{k+1|k+1} = \hat{x}_{k+1|k} + K_{x,k+1}(z_{x,k+1} - \hat{z}_{x,k+1|k})$$

- updated state covariance matrix:

$$P_{k+1|k+1} = P_{k+1|k} - K_{k+1}S_{k+1}K_{k+1}^T$$

C. Controller Design

The benefit of high accuracy in the friction estimation can be clearly visualized when the information is used for compensation by respectable controllers. To arrive at a decent control design, we adopt the idea of the separation principle, i.e., the controller is designed assumed that the state is available, see [20] for an example in a nonlinear system. The controller u consists of the feedforward compensation, u_f and the feedback part, u_b , i.e.,

$$u = u_f + u_b \quad (8)$$

In this study, we propose $u_f = J\ddot{\theta}_d + T_f$, and $u_b = k_d(\dot{\theta}_d - \dot{\theta}) + k_p(\theta_d - \theta)$. By substitute the expression for u in the dynamical equation, we obtain the classical linear error model,

$$J\ddot{e} + k_d\dot{e} + k_p e = 0 \quad (9)$$

where $e = \theta_d - \theta$. By selecting the appropriate values for k_d , and k_p (e.g., Routh-Hurwitz stable), the error would approach zero asymptotically. Note that this may not be the case when the disturbance or the estimation error is sufficiently large. Since the friction and the state are estimated via the EKF or UKF, in the expression (8) T_f becomes $\hat{T}_f$; θ is replaced by $\hat{\theta}$, and $\dot{\theta}$ is substituted by $\dot{\hat{\theta}}$. In the presence of the estimator, the controller now becomes

$$u = J\ddot{\theta}_d + \hat{T}_f + k_d(\dot{\theta}_d - \dot{\hat{\theta}}) + k_p(\theta_d - \hat{\theta}) \quad (10)$$

IV. SIMULATION IMPLEMENTATION

A. Dynamical State Equation

In this study, the real friction is generated based on the Dahl model in which all parameters are given in Table I. The diagram of position control using PD controller, feedforward, and friction compensation with KF (EKF and UKF)-based estimation is shown in Fig. 1.

In the friction estimation part, EKF and UKF are used with the Coulomb and the Dahl friction models. Based on the forward-Euler method, the Coulomb and the Dahl friction models can be written as follows:

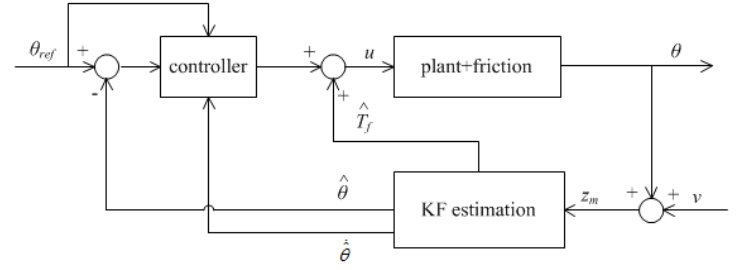


Fig. 1. The diagram of position control using PD controller and friction compensation with KF-based estimation.

1) Coulomb model:

$$x_{1,k+1} = x_{1,k} + \frac{T}{J}(u_k - x_{3,k}\text{sgn}(x_{1,k})) \quad (11)$$

$$x_{2,k+1} = x_{2,k} + T x_{1,k} \quad (12)$$

$$x_{3,k+1} = x_{3,k} + w_k \quad (13)$$

where $x_{1,k} = \dot{\theta}_k, x_{2,k} = \theta_k, x_{3,k} = T_{c,k}$, and T is sampling time. Note that sgn function can lead to a divergent problem. Therefore, in this paper, tanh, a smooth approximation, is used. The measurement equation is

$$z_{k+1} = x_{2,k+1} + v_{k+1}. \quad (14)$$

The Jacobian matrices of the Coulomb model given as follows:

$$F_k = \begin{pmatrix} f_{11} & 0 & f_{13} \\ T & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, H_{k+1} = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix} \quad (15)$$

where $f_{11} = 1 - \frac{T}{J}\text{sech}^2(x_{1,k})$, and $f_{13} = -\frac{T}{J}\tanh(x_{1,k})$.

2) Dahl model:

$$x_{1,k+1} = x_{1,k} + \frac{T}{J}(u_k - x_{3,k}) \quad (16)$$

$$x_{2,k+1} = x_{2,k} + T x_{1,k} \quad (17)$$

$$x_{3,k+1} = x_{3,k} + \sigma T \left(x_{1,k} - \frac{x_{3,k}}{T_c} x_{1,k} \right) + w_k \quad (18)$$

where $x_{1,k} = \dot{\theta}_k, x_{2,k} = \theta_k, x_{3,k} = T_{f,k}$, and the Jacobian matrix is given as

$$F_k = \begin{pmatrix} 1 & 0 & -\frac{T}{J} \\ T & 1 & 0 \\ \sigma T \left(1 - \frac{x_{3,k}}{T_c} \right) & 0 & 1 + \sigma T \left(-\frac{x_{1,k}}{T_c} \right) \end{pmatrix}. \quad (19)$$

3) Controller: We explore the effect of the friction compensation by comparing two versions of the controller designed in the previous section of this paper. The baseline result is obtained by having the estimated $\hat{T}_f$ set to zero in one of the controllers, i.e., the friction is uncompensated. The performance is then compared against the one by the controller in (10) with the compensated friction. The controller is implemented as

$$u_k = J\ddot{\theta}_{d,k} + \hat{x}_{3,k} + k_d(\dot{\theta}_{d,k} - \dot{\hat{x}}_{1,k}) + k_p(\theta_{d,k} - \hat{x}_{2,k}) \quad (20)$$

TABLE II
COMPARISON OF NRMSE BETWEEN REFERENCE AND ESTIMATED POSITION
VALUES

Friction compensation model	Position tracking error
none	0.3044
Coulomb+EKF	0.3161
Coulomb+UKF	0.1101
Dahl+EKF	0.2892
Dahl+UKF	0.0136

4) Parameter setting:

- initial condition of state $x = 0.1 * (1 \ 1 \ 1)^T$ ($[-]^T$ is the matrix transpose).
- initial condition of state covariance matrix (P) is set to $I_{3 \times 3}$, where I is an identity matrix.
- process noise covariance (M) is set to 0.001 and measurement noise covariance (R) is set to 0.01.
- the amplitude of the sinusoidal input is set to 1, with frequency of 0.1 Hz. T is fixed at 0.0001.
- UKF parameters are $\alpha = 1$, $\beta = 2$, $\kappa = 0$.
- the plant's closed loop poles are at $20 \pm 10j$; hence, $k_d = 40 \cdot J$, and $k_p = 500 \cdot J$ in the controller.

B. Simulation Results

The tracking performance is compared in terms of normalized root mean square error (nrmse). The nrmse can be computed as

$$\text{nrmse} = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (r_i - y_i)^2}}{\sqrt{\frac{1}{N} \sum_{i=1}^N (r_i)^2}} \quad (21)$$

where N is a number of data points, r and y are the desired position and the output, respectively. The nrmse between the reference and the estimated position using PD controller with and without friction compensation are depicted in Table II.

Table II compares the tracking errors between the PD controller with and without friction compensation. The UKF based estimation with the Dahl model gives the lowest error, whereas the EKF based estimation with the Coulomb model is the highest. In addition, it is found that the UKF based controllers deliver superior performance to EKF for both with the Coulomb and the Dahl models. Ninety-five percent of the error is reduced for the UKF with the Dahl model and 63% for the UKF with the Coulomb model comparing to the performance by the controller without friction compensation. In contrast, the EKF with the Coulomb model gives the worst performance. Similarly the tracking error of the EKF with the Dahl model is reduced only 5% compared with no friction compensation controller. This might result from the fact that the EKF technique cannot estimate the friction in the nonlinear system very well. Hence, the friction has not truly been compensated.

This is also confirmed by the clear offset in the desired and the actual trajectory of the EKF based controllers shown in Fig. 2. Clearly the linearization of the friction model is

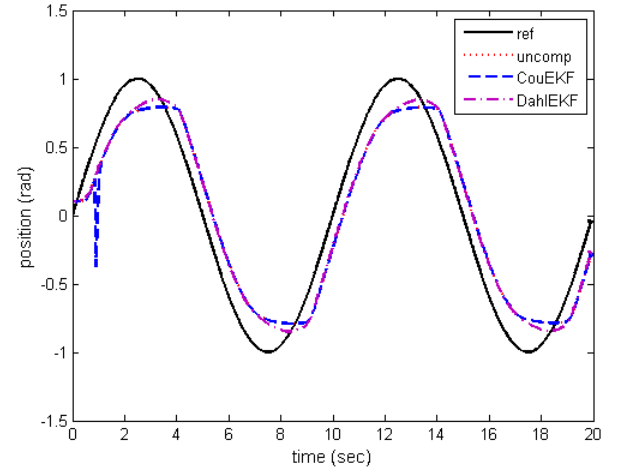


Fig. 2. Comparison of the reference position, without and with friction compensation using EKF for Coulomb and Dahl models .

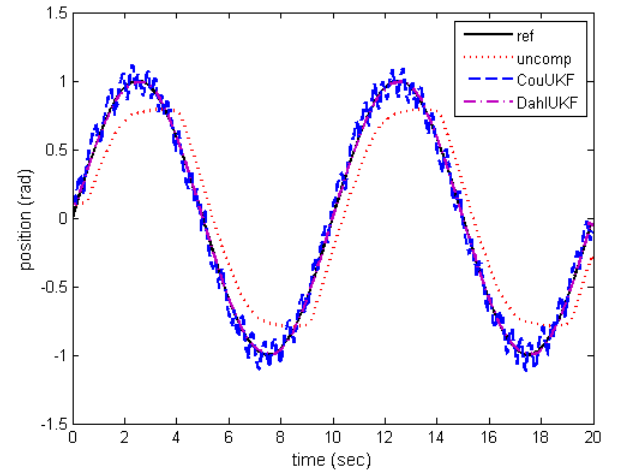


Fig. 3. Comparison of the reference position, without and with friction compensation using UKF for Coulomb and Dahl models.

not sufficiently accurate. On the other hand, both UKF based with the Dahl and the Coulomb models can track the desired trajectory as illustrated in Fig. 3.

V. CONCLUSION

Model-based friction estimation using EKF and UKF with the Coulomb and the Dahl friction models were studied and used to estimate the nonlinear system friction. Based on the simulation results and the real Dahl friction, it was found that UKF with the Dahl model provided the best friction estimation. Also, it can be said that the UKF is superior to the EKF technique. The linearization in the EKF estimator is inaccurate when the friction is highly nonlinear. In contrast, accurate estimation, as in the UKF with the Dahl model can be used for compensation and result in the error approaching zero asymptotically.

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